

• 算法及硬件评估——Ozaki Scheme II

Ozaki Scheme II: A GEMM-oriented emulation of floating-point matrix multiplication using an integer modular technique: 将FP64转换为INT8计算

Algorithm 1 Outline of proposed method using INT8 tensor cores. The functions $\text{trunc}(\cdot)$ and $\text{round}(\cdot)$ round inputs into integers in round-towards-zero and round-to-nearest-even mode, and E is all-ones matrix of appropriate size. $\mathbb{F}_{64}$ is a set of floating-point numbers in FP64.

Input: $A \in \mathbb{F}_{64}^{p \times q}$, $B \in \mathbb{F}_{64}^{q \times r}$, $m \in \mathbb{N}^s$, $s \in \mathbb{N}$

Convert FP64 to INT8:

- 1: Determine shift values $D \in \mathbb{F}_{64}^{p \times q}$ and $E \in \mathbb{F}_{64}^{q \times r}$
- 2: $A' := \text{trunc}(DA)$ $\{A' \in \mathbb{F}_{64}^{p \times q} \cap \mathbb{Z}^{p \times q}\}$
- 3: $B' := \text{trunc}(BE)$ $\{B' \in \mathbb{F}_{64}^{q \times r} \cap \mathbb{Z}^{q \times r}\}$
- 4: $(a'_t)_{ij} := a'_{ij} \bmod m_t$ ($1 \leq t \leq s$)
- 5: $(b'_t)_{ij} := b'_{ij} \bmod m_t$ ($1 \leq t \leq s$)

Matrix multiplication using INT8 TCs:

- 6: $C'_t := A'_t B'_t$ ($1 \leq t \leq s$)

Convert matrix products into UINT8:

- 7: $(c''_t)_{ij} := (c'_t)_{ij} - \lfloor (c'_t)_{ij} / m_t \rfloor m_t$ ($1 \leq t \leq s$)

Accumulate matrix products and inversely scale:

- 8: $C''' := \sum_{t=1}^s C''_t \cdot My_t / m_t$
- 9: $C''' := C''' \bmod M$
- 10: $C := D^{-1} C''' E^{-1}$

Output: $C \in \mathbb{F}_{64}^{p \times r}$

Line1-3: 缩放, 将浮点转为整数

类似于CIM中的前预对齐,

s=16个模, Ka=Kb=53bit就可以表示FP64尾数(有优化空间)

Line4-5: 取模转换为INT8 (常数取模有优化空间)

Line6: INT8矩阵乘法(有优化空间)

Line7: 矩阵乘累加结果继续对常数取模

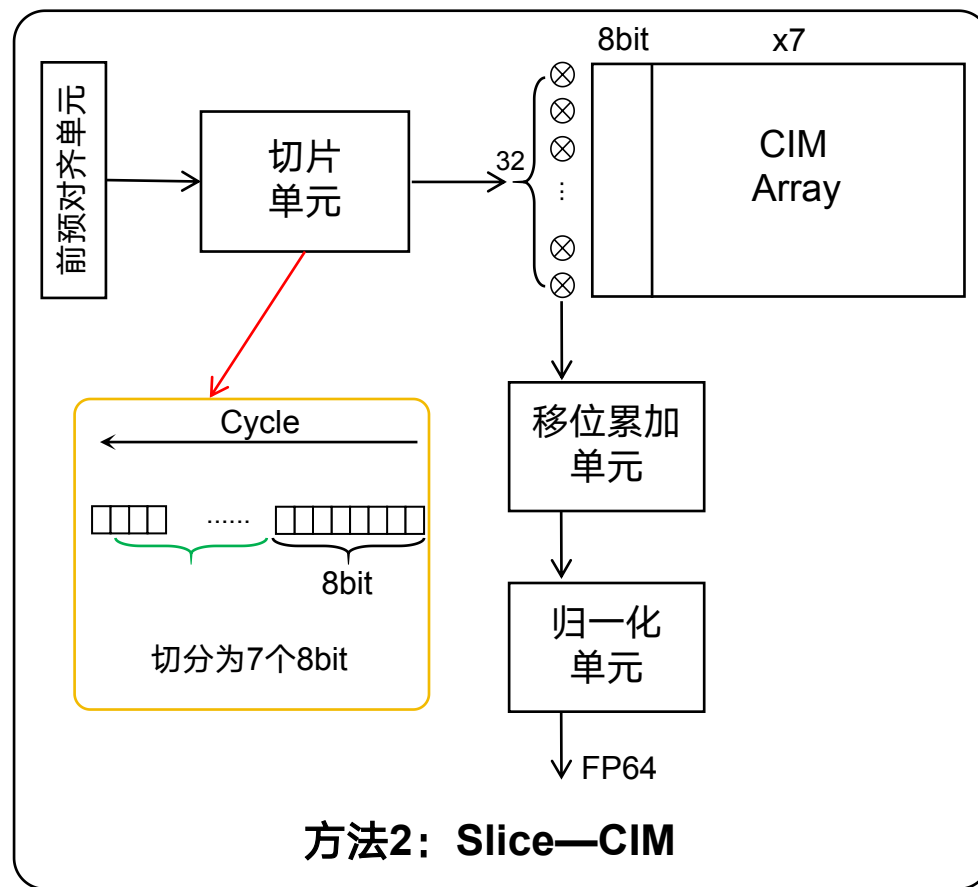
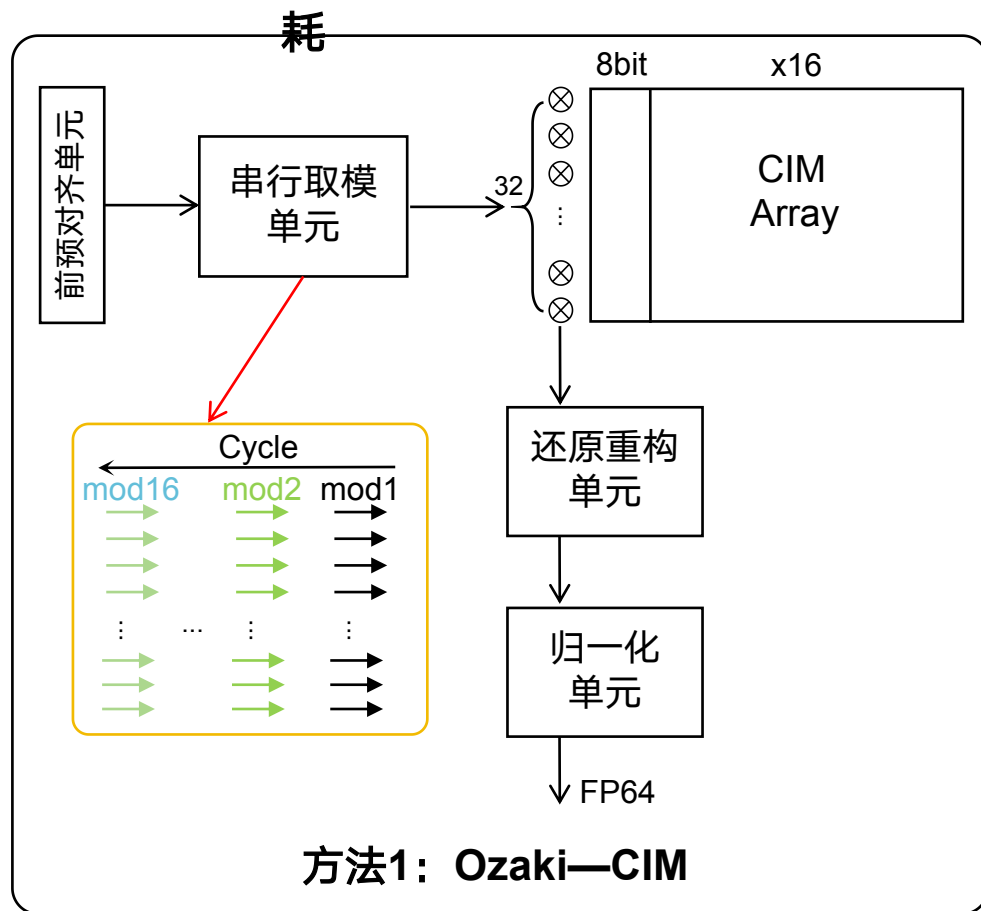
Line8-9: 还原重构(中国剩余定理/Garner method)(有优化空间)

Line10: 归一化为FP64

• 算法及硬件评估——Ozaki Scheme II

评估标准

计算32个FP64 Input和32个 FP64 Weight的乘累加，比较计算周期，面积和功



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算法验证

Python构建两种方法的计算模型

计算结果一致

```
[Ozaki Scheme II]
原始整数结果: 151675699828736311612005204624904
转换FP64结果: 3.063066364593341e+04
绝对误差: 3.637979e-12
相对误差: 1.187692e-16
```

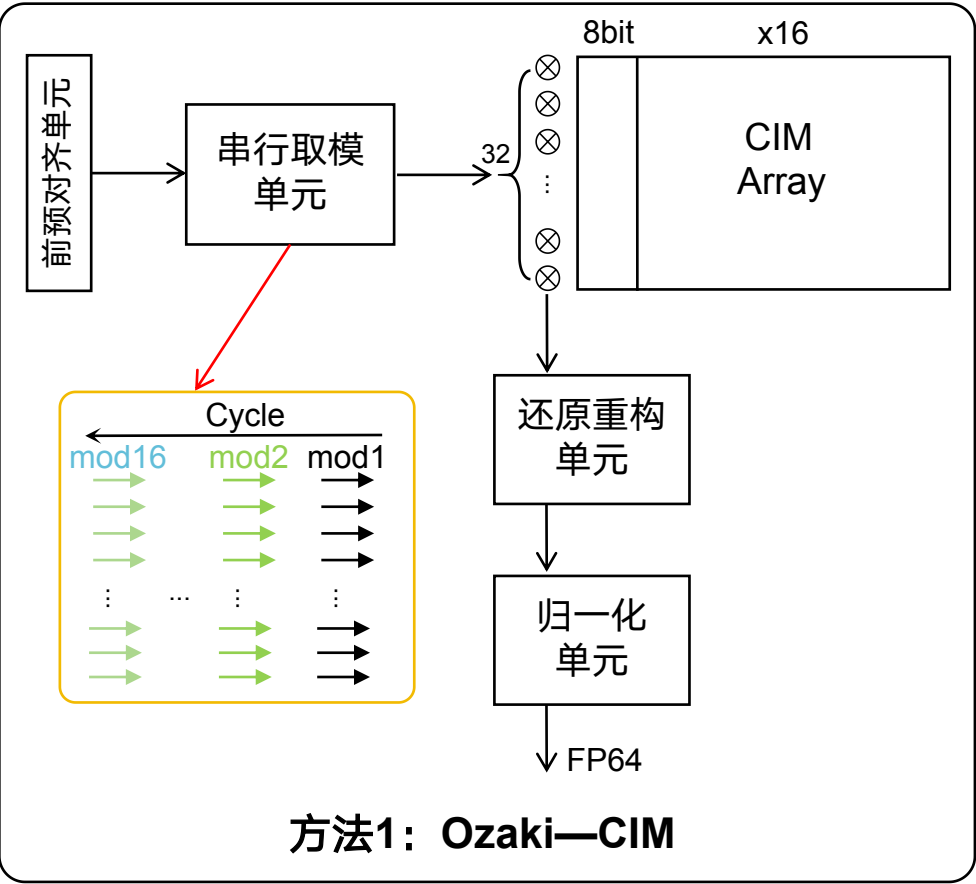
```
[Direct Bit-Slicing]
原始整数结果: 151675699828736311612005204624904
转换FP64结果: 3.063066364593341e+04
绝对误差: 3.637979e-12
相对误差: 1.187692e-16
```

统计结果:
Ozaki - 平均: 5.69e-16, 最大: 1.80e-14
Direct - 平均: 5.69e-16, 最大: 1.80e-14

Test 1: Ozaki=3.37e-16, Direct=3.37e-16	Test 51: Ozaki=0.00e+00, Direct=0.00e+00
Test 2: Ozaki=1.34e-16, Direct=1.34e-16	Test 52: Ozaki=1.88e-16, Direct=1.88e-16
Test 3: Ozaki=1.87e-16, Direct=1.87e-16	Test 53: Ozaki=0.00e+00, Direct=0.00e+00
Test 4: Ozaki=2.50e-16, Direct=2.50e-16	Test 54: Ozaki=3.35e-16, Direct=3.35e-16
Test 5: Ozaki=1.82e-16, Direct=1.82e-16	Test 55: Ozaki=1.80e-14, Direct=1.80e-14
Test 6: Ozaki=1.96e-16, Direct=1.96e-16	Test 56: Ozaki=0.00e+00, Direct=0.00e+00
Test 7: Ozaki=0.00e+00, Direct=0.00e+00	Test 57: Ozaki=0.00e+00, Direct=0.00e+00
Test 8: Ozaki=8.24e-15, Direct=8.24e-15	Test 58: Ozaki=1.68e-16, Direct=1.68e-16
Test 9: Ozaki=0.00e+00, Direct=0.00e+00	Test 59: Ozaki=0.00e+00, Direct=0.00e+00
Test 10: Ozaki=2.12e-16, Direct=2.12e-16	Test 60: Ozaki=2.16e-16, Direct=2.16e-16
Test 11: Ozaki=0.00e+00, Direct=0.00e+00	Test 61: Ozaki=1.47e-16, Direct=1.47e-16
Test 12: Ozaki=3.93e-16, Direct=3.93e-16	Test 62: Ozaki=1.90e-16, Direct=1.90e-16
Test 13: Ozaki=3.21e-16, Direct=3.21e-16	Test 63: Ozaki=1.64e-16, Direct=1.64e-16
Test 14: Ozaki=8.66e-16, Direct=8.66e-16	Test 64: Ozaki=0.00e+00, Direct=0.00e+00
Test 15: Ozaki=0.00e+00, Direct=0.00e+00	Test 65: Ozaki=0.00e+00, Direct=0.00e+00
Test 16: Ozaki=5.84e-16, Direct=5.84e-16	Test 66: Ozaki=2.08e-16, Direct=2.08e-16
Test 17: Ozaki=3.43e-16, Direct=3.43e-16	Test 67: Ozaki=2.79e-16, Direct=2.79e-16
Test 18: Ozaki=1.67e-16, Direct=1.67e-16	Test 68: Ozaki=2.20e-16, Direct=2.20e-16
Test 19: Ozaki=2.21e-16, Direct=2.21e-16	Test 69: Ozaki=0.00e+00, Direct=0.00e+00
Test 20: Ozaki=1.52e-16, Direct=1.52e-16	Test 70: Ozaki=1.40e-16, Direct=1.40e-16
Test 21: Ozaki=1.63e-15, Direct=1.63e-15	Test 71: Ozaki=3.44e-16, Direct=3.44e-16
Test 22: Ozaki=4.77e-16, Direct=4.77e-16	Test 72: Ozaki=1.27e-16, Direct=1.27e-16
Test 23: Ozaki=1.47e-16, Direct=1.47e-16	Test 73: Ozaki=2.09e-16, Direct=2.09e-16
Test 24: Ozaki=0.00e+00, Direct=0.00e+00	Test 74: Ozaki=8.09e-16, Direct=8.09e-16
Test 25: Ozaki=1.16e-16, Direct=1.16e-16	Test 75: Ozaki=0.00e+00, Direct=0.00e+00
Test 26: Ozaki=1.28e-16, Direct=1.28e-16	Test 76: Ozaki=0.00e+00, Direct=0.00e+00
Test 27: Ozaki=0.00e+00, Direct=0.00e+00	Test 77: Ozaki=0.00e+00, Direct=0.00e+00
Test 28: Ozaki=0.00e+00, Direct=0.00e+00	Test 78: Ozaki=1.26e-16, Direct=1.26e-16
Test 29: Ozaki=1.93e-16, Direct=1.93e-16	Test 79: Ozaki=1.65e-16, Direct=1.65e-16
Test 30: Ozaki=3.42e-16, Direct=3.42e-16	Test 80: Ozaki=4.58e-16, Direct=4.58e-16
Test 31: Ozaki=0.00e+00, Direct=0.00e+00	Test 81: Ozaki=0.00e+00, Direct=0.00e+00
Test 32: Ozaki=1.94e-16, Direct=1.94e-16	Test 82: Ozaki=3.11e-16, Direct=3.11e-16
Test 33: Ozaki=3.20e-16, Direct=3.20e-16	Test 83: Ozaki=1.12e-15, Direct=1.12e-15
Test 34: Ozaki=1.58e-15, Direct=1.58e-15	Test 84: Ozaki=1.29e-15, Direct=1.29e-15
Test 35: Ozaki=0.00e+00, Direct=0.00e+00	Test 85: Ozaki=1.67e-16, Direct=1.67e-16
Test 36: Ozaki=1.15e-16, Direct=1.15e-16	Test 86: Ozaki=1.89e-16, Direct=1.89e-16
Test 37: Ozaki=1.22e-16, Direct=1.22e-16	Test 87: Ozaki=1.62e-16, Direct=1.62e-16
Test 38: Ozaki=1.88e-16, Direct=1.88e-16	Test 88: Ozaki=1.25e-16, Direct=1.25e-16
Test 39: Ozaki=2.01e-16, Direct=2.01e-16	Test 89: Ozaki=1.26e-15, Direct=1.26e-15
Test 40: Ozaki=3.77e-16, Direct=3.77e-16	Test 90: Ozaki=4.40e-15, Direct=4.40e-15
Test 41: Ozaki=0.00e+00, Direct=0.00e+00	Test 91: Ozaki=1.08e-15, Direct=1.08e-15
Test 42: Ozaki=7.21e-16, Direct=7.21e-16	Test 92: Ozaki=0.00e+00, Direct=0.00e+00
Test 43: Ozaki=0.00e+00, Direct=0.00e+00	Test 93: Ozaki=1.36e-16, Direct=1.36e-16
Test 44: Ozaki=5.65e-16, Direct=5.65e-16	Test 94: Ozaki=2.14e-16, Direct=2.14e-16
Test 45: Ozaki=9.01e-16, Direct=9.01e-16	Test 95: Ozaki=1.39e-16, Direct=1.39e-16
Test 46: Ozaki=0.00e+00, Direct=0.00e+00	Test 96: Ozaki=2.87e-16, Direct=2.87e-16
Test 47: Ozaki=1.51e-16, Direct=1.51e-16	Test 97: Ozaki=1.90e-16, Direct=1.90e-16
Test 48: Ozaki=1.14e-16, Direct=1.14e-16	Test 98: Ozaki=1.59e-15, Direct=1.59e-15
Test 49: Ozaki=0.00e+00, Direct=0.00e+00	Test 99: Ozaki=0.00e+00, Direct=0.00e+00
Test 50: Ozaki=4.75e-16, Direct=4.75e-16	Test 100: Ozaki=0.00e+00, Direct=0.00e+00

• 算法及硬件评估——Ozaki Scheme II

硬件评估



串行取模单元：输入串行对已知常数取模

$$m := (256, 255, 253, 251, 247, 239, 233, 229, 227, 223, 217, 211, 199, 197, 193, 191)^T \in \mathbb{N}^{16}.$$

- 1、最直接的就是 $X \bmod m$ ，但硬件代价大(有除法器)
- 2、对常数取模进行一定优化

$$X = B_6 \cdot (2^8)^6 + B_5 \cdot (2^8)^5 + B_4 \cdot (2^8)^4 + B_3 \cdot (2^8)^3 + B_2 \cdot (2^8)^2 + B_1 \cdot (2^8)^1 + B_0$$

$$2^8 = 256 \equiv k \pmod{m}$$

$$X \pmod{m} = (B_6 \cdot k^6 + B_5 \cdot k^5 + B_4 \cdot k^4 + B_3 \cdot k^3 + B_2 \cdot k^2 + B_1 \cdot k + B_0) \pmod{m}$$

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索引 t	模数 m_t	基数 k	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
		$(256 - m_t)$	(k^0)	(k^1)	(k^2)	(k^3)	(k^4)	(k^5)	(k^6)
0	256	0	1	0	0	0	0	0	0
1	255	1	1	1	1	1	1	1	1
2	253	3	1	3	9	27	81	243	223
3	251	5	1	5	25	125	123	113	63
4	247	9	1	9	81	235	139	16	144
5	239	17	1	17	50	133	110	197	3
6	233	23	1	23	63	51	8	184	38
7	229	27	1	27	42	218	161	225	121
8	227	29	1	29	160	100	176	110	12
9	223	33	1	33	197	34	7	8	41
10	217	39	1	39	2	78	4	156	8
11	211	45	1	45	126	184	51	185	96
12	199	57	1	57	65	123	46	35	5
13	197	59	1	59	132	105	88	70	190
14	193	63	1	63	109	112	108	49	192
15	191	65	1	65	23	158	147	5	134

串行取模单元：输入串行对已知常数取模

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$$2^8 = 256 \equiv k \pmod{m}$$

$$X \pmod{m} \equiv (B_6 \cdot k^6 + B_5 \cdot k^5 + B_4 \cdot k^4 + B_3 \cdot k^3 + B_2 \cdot k^2 + B_1 \cdot k + B_0) \pmod{m}$$

把大代价的除法转为小代价的乘和加

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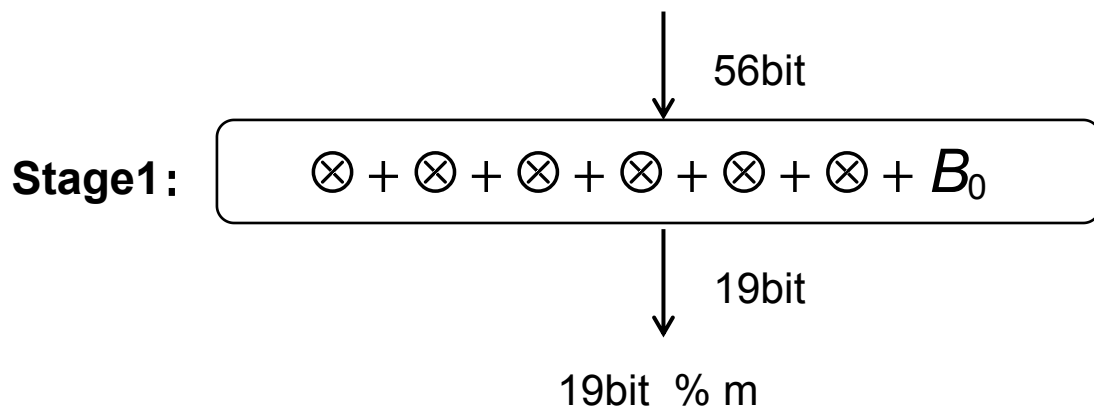
对常数取模进行一定优化

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$$2^8 = 256 \equiv k \pmod{m}$$

$$X \pmod{m} \equiv (B_6 \cdot k^6 + B_5 \cdot k^5 + B_4 \cdot k^4 + B_3 \cdot k^3 + B_2 \cdot k^2 + B_1 \cdot k + B_0) \pmod{m}$$

把大代价的除法转为小代价的乘和加



1、直接%的面积：1073

```
Total cell area: 1073.986550
Total area: undefined
1
```

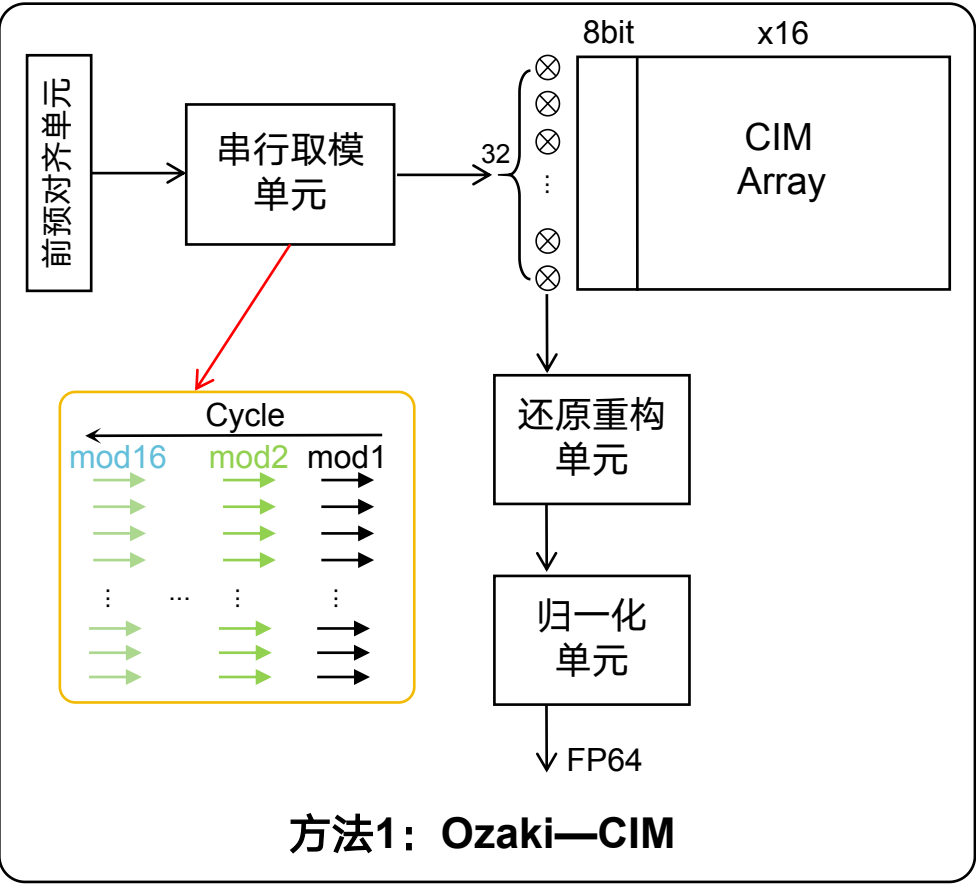
2、优化后的面积：761

```
Total cell area: 761.932800
Total area: undefined
1
```

面积减少29%

• 算法及硬件评估——Ozaki Scheme II

硬件评估



还原重构单元：将余数结果还原为大整数

1、原文中的中国剩余定理

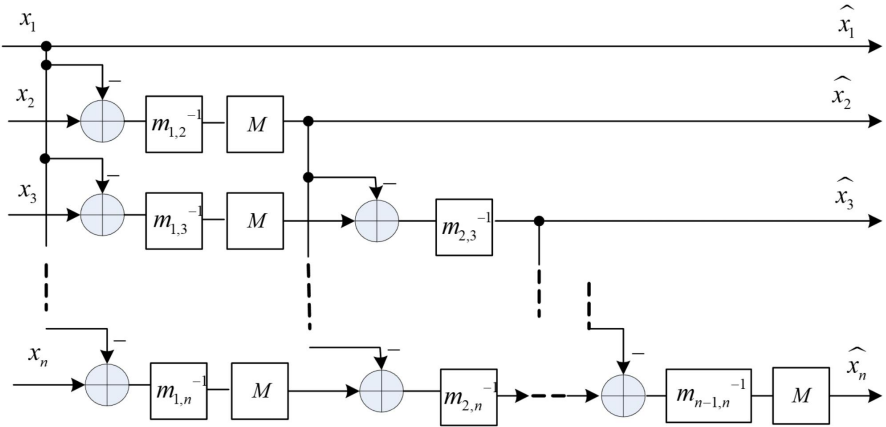
$$\begin{aligned} 8: C''' &:= \sum_{t=1}^s C_t'' \cdot My_t/m_t \\ 9: C''' &:= C''' \bmod M \end{aligned}$$

代价非常大：超长位宽乘法(128bit)，超长位宽取模

2、混合基转换

$$X = v_1 \hat{x}_1 + v_2 \hat{x}_2 + \dots + v_n \hat{x}_n$$

其中， $v_1 = 1$ ， $v_2 = m_1$ ， $v_3 = m_1 m_2$ ， $v_4 = m_1 m_2 m_3$ ， $\dots$

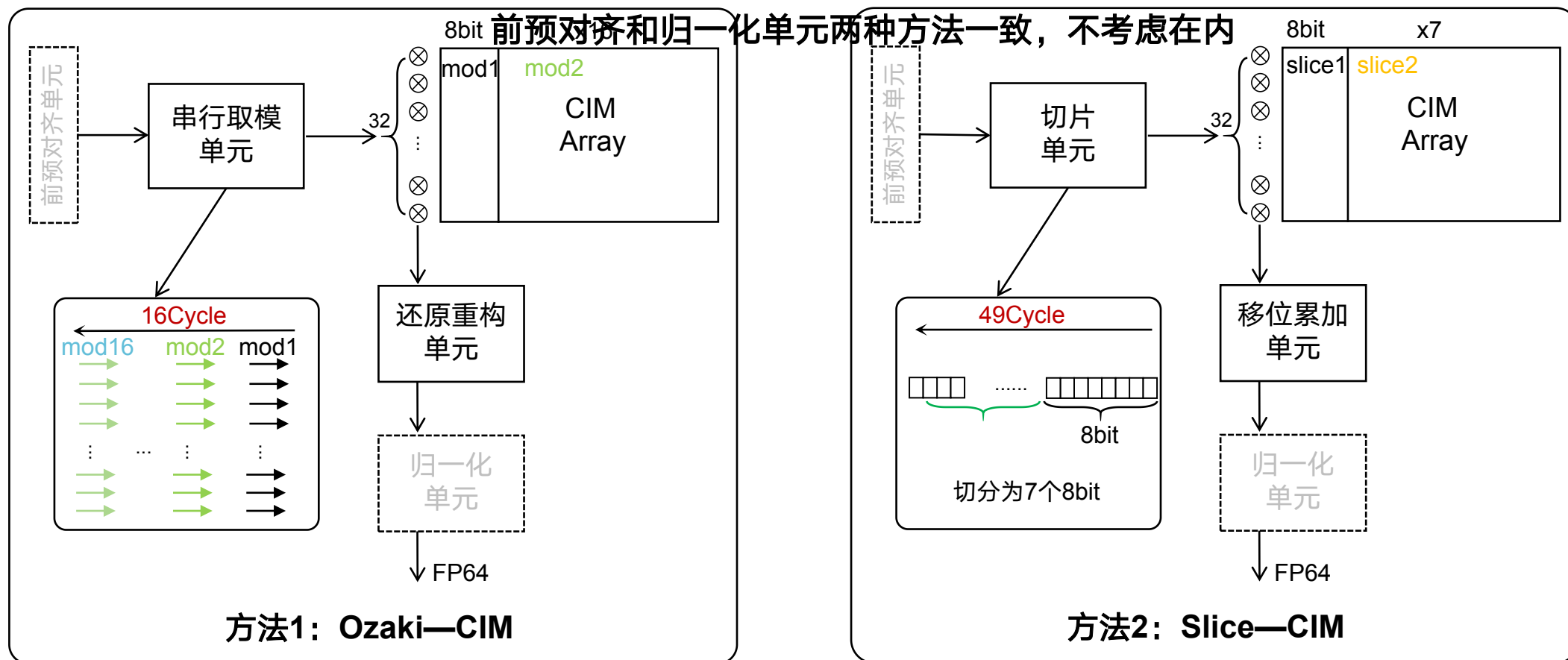


- 算法及硬件评估——Ozaki Scheme II

硬件评估

计算32个FP64 Input和32个 FP64 Weight的乘累加，比较计算周期，面积和功

耗



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硬件评估

1、计算周期： Ozaki—CIM : 16周期 Slice—CIM: 49周期 $49/16=3.06$

2、面积： Ozaki—CIM : 32576 Slice—CIM: 4402 $32576/4402=7.4$

Hierarchical cell	Absolute
	Total
-----	-----
ozaki_top	32576.3479
u_cim_core	2136.1766
u_mrc	4749.4655
u_serial_modulo	25434.2245
-----	-----
Total	

Hierarchical cell	Absolute
	Total
-----	-----
direct_top	4402.0684
u_bit_slicer	625.9968
u_cim_core	2785.1673
u_shift_acc	730.9209
-----	-----
Total	

大部分面积为串行取模单元，32行并行取模单元

虽然面积较大，但作为全局处理单元可被所有macro共享，平摊开销

- 算法及硬件评估——Ozaki Scheme II

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3、功耗： Ozaki—CIM : 19.67mw Slice—CIM: 20.874mw

模块	Switch Power	Internal Power	Leakage	Total	占比
u_serial_modulo	3.114 mW	3.475 mW	8.62 μ W	6.598 mW	88.2%
u_mrc	0.271 mW	0.276 mW	12.4 μ W	0.559 mW	7.5%
u_cim_core	0.125 mW	0.133 mW	0.15 μ W	0.258 mW	3.5%
Total	3.517 mW	3.943 mW	21.3 μ W	7.481 mW	100%

模块	Switch Power	Internal Power	Leakage	Total	占比
u_cim_core	0.085 mW	0.098 mW	0.20 μ W	0.183 mW	43.1%
u_shift_acc	0.043 mW	0.063 mW	10.1 μ W	0.117 mW	27.4%
u_bit_slicer	0.003 mW	0.059 mW	0.04 μ W	0.062 mW	14.5%
Total	0.134 mW	0.281 mW	10.5 μ W	0.426 mW	100%

大部分功耗为串行取模单元，32行并行取模单元

虽然功耗较大，但作为全局处理单元计算一次的结果可被多次共享，平摊开销

• 算法及硬件评估——Ozaki Scheme II

硬件评估

4、拓展： 直接实现56bit*56bit的乘累加

面积： Ozaki—CIM : 32576 Slice—CIM: 4402
56bit*56bit: 81280

功耗： Ozaki—CIM : 19.67mw Slice—CIM: 20.874mw
56bit*56bit: 26.89mw

		Global cell area	

Hierarchical cell		Absolute	Perce
		Total	Total
-----		-----	-----
mul56_top		81280.0041	100
-----		-----	-----
Total			
1			

Total Dynamic Power = 26.8796 mW (100%)					
Cell Leakage Power = 15.8209 uW					
Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)

io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)
clock_network	5.7968e-02	0.0000	0.0000	0.0000	(0.00%)
register	9.3264e-03	5.4601e-03	8.5785	7.2763e-02	(0.27%)
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)
combinational	12.1393	14.6703	1.5812e+04	26.8227	(99.73%)

Total	12.2066 mW	14.6758 mW	1.5821e+04 nW	26.8954 mW	

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